

Breaking the Ice with Group Flow: A Collaborative VR Serious Game for Relationship Enhancement

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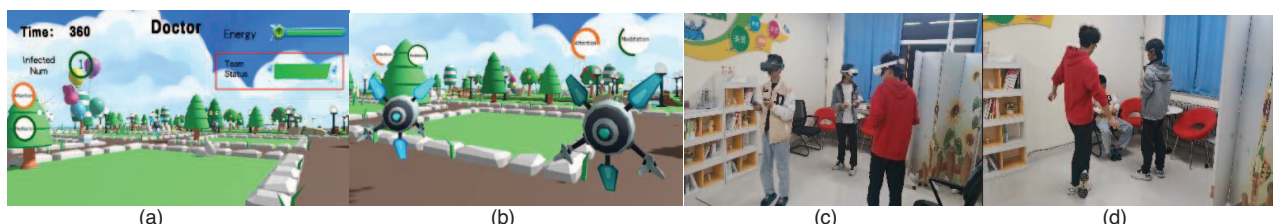
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Figure 1: The collaborative VR serious game prototype for relationship enhancement: (a) a snapshot of the game interface from first-person perspective, including dynamic team flow level, completion progress of teamwork, and visualized personal mental state; (b) visualization and sharing of teammates' mental states; (c) a snapshot of the user study, showing three participants experiencing the VR serious game as a team; (d) teammates are communicating and discussing the strategy to overcome the difficulty in the task.

ABSTRACT

It is usually hard for unfamiliar partners to rapidly “break the ice” in the early stage of relationship establishment, which hinders the development of relationship and even affects the team productivity. To solve this problem, we proposed a collaborative serious game for icebreaking by combining immersive virtual reality (VR) with brain-computer interface based on the team flow framework. We designed a multiplayer collaboration task with the theme of fighting COVID-19 and proposed an approach to improve empathy between team members by sharing their real-time mental state in VR; in addition, we propose an EEG-based method for dynamic evaluation and enhancement of group flow experience to achieve better team collaboration. Then, we developed a prototype system and performed a user study. Results show that our method has good ease of use and can significantly reduce the psychological distance among team members. Especially for unfamiliar partners, both functions of mental state sharing and group flow regulation enhancement can significantly reduce the psychological distance.

Keywords: Group flow, multi-player collaboration, brain-computer interaction, VR serious game.

Index Terms: Human-centered computing—Ubiquitous and mobile computing—Ubiquitous and mobile computing systems and tools

1 INTRODUCTION

Teamwork is an important way to develop social relationships and achieve team goals. For unfamiliar teammates in a group, it is

difficult to “break the ice” in the early stages of developing social relationship, which may affect the teamwork performance and experience. Utilizing new technologies/methods to enhance human relationships and performance in teamwork is an important topic of VR community. Collaborative VR serious game maybe a potential approach of icebreaking. VR can help people be immersed in the social situation and enhance the emotional satisfaction [1]. Cooperative VR serious games help enhance the emotions among teammates and team performance. However, most existing VR serious games have the following issues: 1) the current VR games isolate VR environment from physical world. This makes it difficult to understand the teammates' state and empathize with them, further leading to missing opportunity for necessary communication; 2) although the assessment of user experience is increasingly focused in individual level, there is still a lack of real-time method to assess user experience in group level during cooperative teamwork [2]. These issues may weaken the positive effect of collaborative VR serious game on relationship enhancement.

To address these issues, this study focuses on the concept of team flow. Group flow is a shared state of balance within a group/team as represented by (a) fluent, positive interactions within the group, (b) a high collective competence of the group and (c) a collective mind state of group by means of positive relationships among group members, often resulting in optimal collective performance, creativity, and positive collective experience. [3] Reaching flow state in individual level would promote personal experience and performance; reaching it in a group level would make everyone feel better overall, contributing to the enhancement of common team emotions and collective effectiveness [4]. Therefore, achieving individual flow and group flow is important goals for teamwork. Real-time assessment and sharing of individual mental states help to enhance mutual understanding and empathy, while real-time assessment and enhancement of group flow has a very positive impact on team goal achievement. Brain-computer interface (BCI) can provide a potential solution. Previous studies have shown that EEG activity in frontal lobes is correlated with

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flow states [5], and BCI can be used to effectively assess and enhance user experience in real time.

Therefore, this study intends to propose an effective relationship enhancement method for unfamiliar teammates to “break the ice” by combining VR serious game with BCI. Based on the group flow framework and advantages of VR serious games, this study designed a collaborative VR serious game with the theme of fighting COVID-19 and developed a prototype system. It supports multiple users to play different roles to accomplish a teamwork together in an immersive VR environment (Figure 1a). Each teammate’s mental state (concentration and meditation) could be assessed and visualized through an EEG headband to achieve interactive control and empathy enhancement (Figure 1b). Moreover, a dynamic assessment and enhancement method of group flow experience was proposed to achieve a better teamwork state. Finally, we performed a user study (Figure 1c) to verify the effectiveness of our proposed methods in enhancing the relationships among teammates.

2 SYSTEM DESIGN

According to the group flow framework, we designed a collaborative serious game which included four key modules: (1) multiplayer VR- collaborative- task module (Module 1) was designed to provide multiplayer with immersive VR experience and collaborative tasks with a common goal; (2) multimodal interactive control module (Module 2) was designed to provide behavioral interaction and BCI to achieve natural interaction experience; (3) mental state dynamic visualization module (Module 3) was designed to share teammates’ real-time mental state to achieve better understanding and sympathy with each other; (4) team flow real-time evaluation and adjustment module (Module 4) was designed to dynamically evaluate the group flow level of the team, and adjust the game difficulty to achieve better teamwork experience.

3 SYSTEM IMPLEMENTATION

On the whole, the hardware of our system includes Pico Neo 3 and BrainLink Pro EEG headband; the software includes Unity 3D engine and Photon online plugin. The implementation of each module is briefly introduced as follows.

Implementation of Module 1: the COVID-19 fighting game requires multiple players work as a team to eliminate the virus and cure all patients in a virtual village. The players play the role of service robots with different duties and functions (e.g., doctor, virus tester et al.). They need to (1) create and operate different anti-epidemic (e.g., hospital, virus-testing station), and (2) actively search for severe patients and take them to hospital. Only when all infected villagers are cured within a limited time, can the team goal be achieved successfully. We used the Unity Photon online plugin and the official Pico Neo 3 SDK to implement the multiplayer VR collaborative interaction in the Unity 3D engine.

Implementation of Module 2: multimodal interaction included the interactive handle interaction and brain-computer interaction. For the former, Pico Neo 3 official SDK contains the method of using Pico handle for interaction. BrainLink Pro, an low-cost BCI device, was used to collect EEG signals through electrode pads on the forehead, and then calculate the EEG signals using the eSense algorithm of NeuroSky to get the users’ attention and meditation [6]. Pico Neo 3 official SDK contains the method of using Pico handle for interaction. Based on this, we modify the BrainLink Pro official SDK so that it can be used in the VR scene in Android environment to compute the user’s mental state to realize the multimodal interactive control module.

Implementation of Module 3: Based on the above two modules, this module is implemented in the Unity game engine using C#

code, allowing users to observe the degree of attention and meditation of teammates in real time.

Implementation of Module 4: Each player’s EEG signals were collected in real time during the game. Based on their attention and meditation level, we proposed an algorithm to calculate the team flow state, and then dynamically adjust the game difficulty.

4 USER STUDY

A user study was conducted to test the effect of our proposed method on the relationship enhancement of team members. Thirty college students were recruited and divided into 10 groups to participate the study. Half of the participants were familiar and half were unfamiliar with other teammates before the study. We prepared three versions of the experimental system: version 1 is the system introduced in section3; version 2 removed the visualization display of the teammates’ mental state based on version 1; version 3 removed the dynamic assessment and adjustment of group flow based on version 1.

Before the experiment, participants completed a pre-test questionnaire to assess the psychological distance with other teammates. Then, each group of participants experienced each of the three system conditions, and the order of experience was balanced by a Latin square design. After experienced each condition, participants completed a post-test questionnaire including measures of psychological distance, emotional experience, and team flow experience. The results show that our developed VR serious game can effectively enhance the communication among teammates (Figure 1d). Moreover, the function of dynamic group flow assessment and enhancement can significantly reduce the psychological distance among teammates ($p < 0.05$).

5 CONCLUSION

In this work, we design and develop a collaborative VR serious game for relationship enhancement based on team flow framework. The prototype enables a multiplayer brain-computer collaborative method to help unfamiliar teammate “Break the Ice” through cooperative teamwork. A dynamic evaluation method of team flow experience and visualized sharing method of teammates’ mental states are achieved by using low-cost EEG devices. Results of user study show that our method has good ease of use. For unfamiliar partners, both functions of sharing mental state and group flow enhancement are beneficial to enhancing communication and reducing the psychological distance among teammates.

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